

Partial Algebraic Shifting

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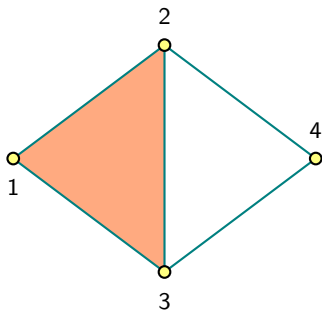
TU Berlin

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Simplicial Complexes

Definition An abstract simplicial complex K , is a collection of non empty sets of $[n] := \{1, 2, \dots, n\}$, such that for any $\tau, \sigma \subset [n]$, if $\tau \subset \sigma \in K$ then $\tau \in K$.

- $\sigma \in K$ a k -simplex or a face, $\dim(\sigma) = k + 1$.
- f -vector number of k -faces for each k .



The f -vector is $(4, 5, 1)$

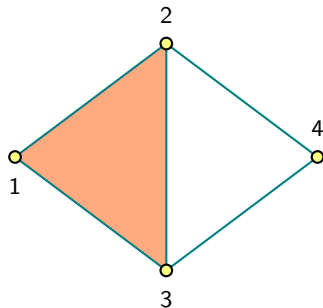
Simplicial Homology

Definition Given a field $\mathbb{F}$, and a simplicial complex K we define the k th simplicial homology to be the vector space

$$H_k^{\mathbb{F}}(K) := k\text{-cycles} / k+1\text{-boundaries}.$$

The k th Betti number of K is the dimension of $H_k^{\mathbb{F}}(K)$.

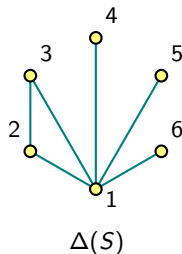
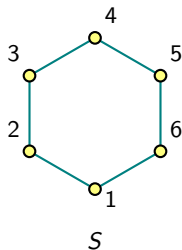
The Betti numbers are $(1, 1, 0)$



Algebraic Shifting

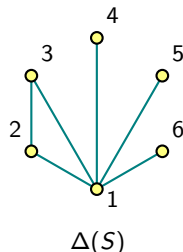
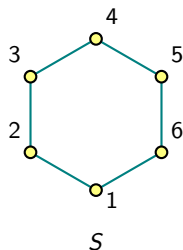
Definition A k -uniform hypergraph is a collection $S \subseteq \binom{[n]}{k}$.

S is shifted if for any $\sigma \in S$, $\sigma|_i^j \in S$ for $i \in \sigma$ and $j < i$.



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Theorem Kalai '84, Björner-Kalai '88 Given K a simplicial complex and a field $\mathbb{F}$, there exists a shifted simplicial complex $\Delta^{\mathbb{F}}(K)$ with the same f -vector and Betti numbers as K .

Exterior Algebraic Shift

Definition Let $\mathbb{E} \supset \mathbb{F}$ be a field extension and let $g \in \mathrm{GL}(n, \mathbb{E})$.

- The k th compound matrix $g^{\wedge k}$ is the $\binom{n}{k} \times \binom{n}{k}$ -matrix of k -minors of g .
- $g^{\wedge S}$ is the $|S| \times \binom{n}{k}$ -submatrix of $g^{\wedge k}$ with rows indexed by S .

Let $\sigma_i \in S$, $|S| = m$ and $\tau_i \in \binom{[n]}{k}$ such that

$$g^{\wedge S} = \begin{pmatrix} \det g_{\sigma_1 \tau_1} & \det g_{\sigma_1 \tau_2} & \cdots & \det g_{\sigma_m \tau_{\binom{n}{k}}} \\ \vdots & \vdots & \vdots & \vdots \\ \det g_{\sigma_m \tau_1} & \det g_{\sigma_m \tau_2} & \cdots & \det g_{\sigma_m \tau_{\binom{n}{k}}} \end{pmatrix}.$$

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The (exterior) algebraic shift is,

$$\Delta_g^{\mathbb{F}}(S) := \left\{ \sigma \in \binom{[n]}{k} : \sigma \text{ indexes a column with a pivot of } g^{\wedge S} \right\}.$$

for a "generic enough" g .

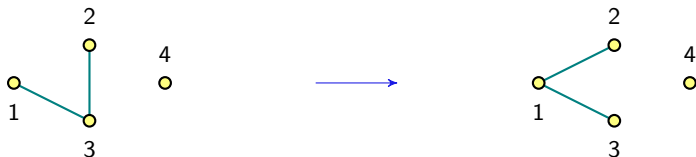
Small Example

For $g = (x_{ij})_{1 \leq i, j \leq 4}$ $S = \{\{1, 2\}, \{2, 3\}\} \subset \binom{[4]}{2}$, $\mathbb{F} = \text{GF}(2)$.

$$g^{\wedge S} = \begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{22}x_{13} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ x_{21}x_{32}+x_{22}x_{31} & x_{21}x_{33}+x_{23}x_{31} & x_{21}x_{34}+x_{24}x_{31} & x_{22}x_{33}+x_{23}x_{32} & x_{22}x_{34}+x_{24}x_{32} & x_{23}x_{34}+x_{24}x_{33} \end{pmatrix}.$$

Which row reduces to,

$$\begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{13}x_{22} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ 0 & x_{11}x_{21}x_{22}x_{33} + x_{11}x_{21}x_{23}x_{32} + x_{12}x_{21}^2x_{33} + x_{13}x_{21}^2x_{32} + x_{12}x_{21}x_{22}x_{31} + x_{13}x_{21}x_{22}x_{31} & x_{11}x_{21}x_{22}x_{34} + x_{11}x_{21}x_{24}x_{32} + x_{12}x_{21}^2x_{34} + x_{14}x_{21}^2x_{32} + x_{12}x_{21}x_{22}x_{31} + x_{12}x_{21}x_{24}x_{31} & x_{11}x_{22}^2x_{33} + x_{11}x_{22}x_{23}x_{32} + x_{12}x_{21}x_{22}x_{33} + x_{13}x_{21}x_{22}x_{32} + x_{12}x_{22}^2x_{31} + x_{13}x_{22}^2x_{31} & x_{11}x_{22}^2x_{34} + x_{11}x_{22}x_{24}x_{32} + x_{12}x_{21}x_{22}x_{34} + x_{14}x_{21}x_{22}x_{32} + x_{12}x_{22}^2x_{31} + x_{12}x_{22}x_{24}x_{31} & x_{11}x_{22}x_{23}x_{34} + x_{11}x_{22}x_{24}x_{33} + x_{12}x_{21}x_{23}x_{34} + x_{12}x_{21}x_{24}x_{33} + x_{13}x_{21}x_{24}x_{32} + x_{14}x_{21}x_{23}x_{32} + x_{13}x_{22}x_{24}x_{31} + x_{14}x_{22}x_{23}x_{31} \end{pmatrix}.$$



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Definition Partial Shift Graph

- V the set of simplicial complexes with a fixed f -vector.
- $\mathcal{F} \subseteq \text{GL}(\mathbb{E}, n)$
- $E = \{(K_1, K_2) \mid \exists M \in \mathcal{F} \text{ with } \Delta_M(K_1) = K_2\}$

The partial shift graph $\Gamma_{\mathcal{F}}$ is the graph with vertex set V and edge set E .

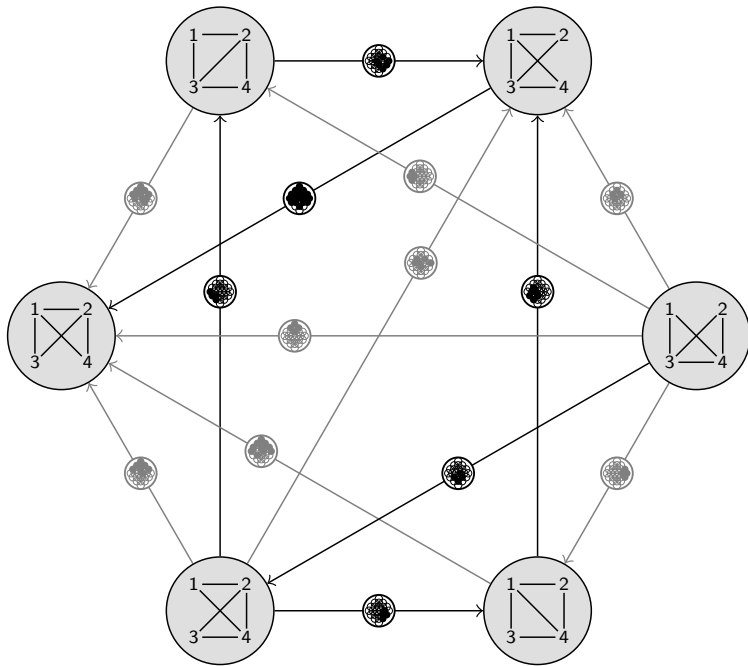
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For which $\mathcal{F}$ is $\Gamma_{\mathcal{F}}$ acyclic?



Matrix decomposition

Lemma $\Delta_{gb}(S) = \Delta_g(S)$ for any upper triangular b and any simplicial complex S .

LU decomposition For $g \in GL(n, \mathbb{E})$ with non zero leading principal minors, there exists l lower triangular, u upper triangular such that

$$g = lu \text{ .}$$

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Bruhat Decomposition Let $B \subset \mathrm{GL}(n, \mathbb{E})$ be the subgroup of upper triangular matrices,

$$\mathrm{GL}(n, \mathbb{E}) = \coprod_{w \in \mathrm{Sym}(n)} BwB .$$

Finding the Bruhat cell of g

Let $g \in GL(\mathbb{E}, n)$ be generic, and let

$$w_0 = (1 \ n)(2 \ n-1) \dots (\lfloor \frac{n}{2} \rfloor \lceil \frac{n}{2} \rceil + 1)$$

Hence $w_0 g$ is also generic and we have

$$\Delta_{w_0 g}(K) = \Delta_g(K) = \Delta(K).$$

We also have lower triangular l , and upper triangular u', u such that,

$$\begin{aligned} w_0 g &= l u = w_0 u' w_0 u \\ g &= u' w_0 u \end{aligned}$$

$g \in B w_0 B$, and $\Delta_{u' w_0}(K) = \Delta(K)$

Inversions

Definition For $w \in \text{Sym}(n)$, let

$$\text{Inv } w := \{(i, j) : i < j, i \cdot w > j \cdot w\}$$

$$\ell(w) := |\text{Inv } w|$$

$$w_0 = \operatorname{argmax}_{w \in \text{Sym}(n)} \ell(w)$$

Example

Take $w = (2 \ 3)$, $\text{Inv } w = \{(2, 3)\}$.

$$\begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = \begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ & 1 & z \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & y & x \\ & 1 & 0 \\ & & 1 \end{pmatrix} .$$

Rothe Matrices

Definition We call $\mathfrak{r}(w) := \mathfrak{u}(w)w$ the Rothe matrix, where

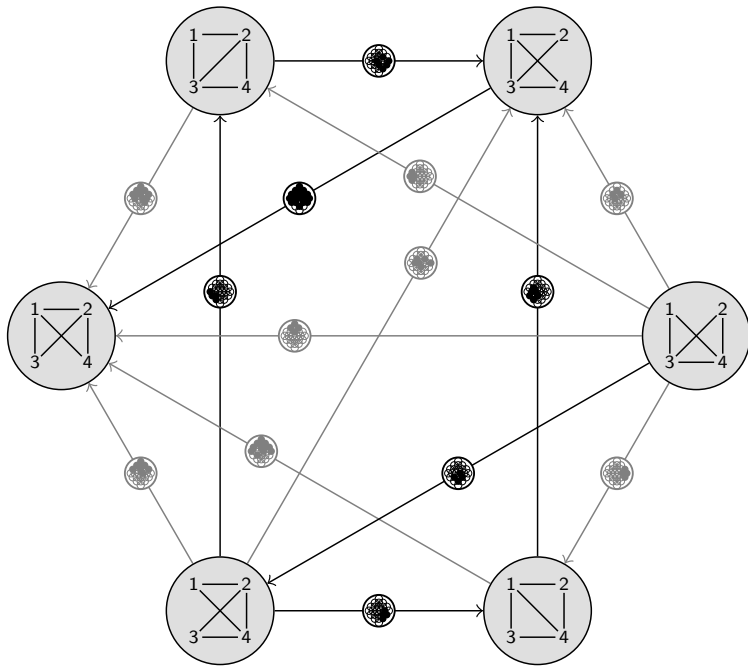
$$\mathfrak{u}(w) := \begin{pmatrix} 1 & u_{12} & \cdots & u_{1n} \\ & \ddots & \ddots & \vdots \\ & & 1 & u_{n-1,n} \\ & & & 1 \end{pmatrix} \quad \text{with} \quad u_{ij} = \begin{cases} x_{ij} & \text{if } (i,j) \in \text{Inv } w, \\ 0 & \text{otherwise} \end{cases}.$$

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Theorem DV-Joswig-Lenzen 24+ For $\mathcal{F} = \{\mathfrak{r}(w) \mid w \in \text{Sym}(n)\}$, $\Gamma_{\mathcal{F}}$ is acyclic.



Weak Bruhat order

Definition Björner-Brenti '05 The weak Bruhat right order is a partial ordering on $\text{Sym}(n)$,

$$w = uv \geq u$$

if $\ell(w) = \ell(u) + \ell(v)$.

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Lemma For $u, v \in \text{Sym}(n)$ with $uv \geq u$ if and only if

$$\text{Inv } uv = \text{Inv } u \cup (\text{Inv } v) \cdot u^{-1}$$

$$\emptyset = \text{Inv } u \cap (\text{Inv } v) \cdot u^{-1}$$

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Example

$$w = s_1 s_2 = (1 \ 3 \ 2), \quad \text{inv}(w) = \{(1, 2), (1, 3)\}$$

$$\text{inv}(s_3) = \{(3, 4)\}$$

$$ws_3 = (1 \ 4 \ 3 \ 2), \quad \text{inv}(ws_3) = \{(1, 2), (1, 3), (1, 4)\}$$

Ingredients to acyclicity proof

Lemma DV-Joswig-Lenzen '24+ Let S be a uniform hypergraph, let $v, w \in \text{Sym}(n)$ with $\ell(vw) = \ell(v) + \ell(w)$ then $\tau(vw)^S$ and $(\tau(v)\tau(w)^v)^{\wedge S}$ have the same column matroid.

Lemma DV-Joswig-Lenzen '24+ Let S be a uniform hypergraph, $w \in \text{Sym}(n)$ and $s = (i \ i+1)$ such that $\ell(ws) = \ell(w) + \ell(s)$,

$$\Delta_{\tau(ws)}(S) \leq \Delta_{\tau(w)}(S) .$$

Theorem DV-Joswig-Lenzen 24+ Let K be a simplicial complex, $c_n = (1 \ 2 \ \cdots \ n)$ and $w \in \text{Sym}(n)$ such that $w \geq c_n$ in the weak order.

- $\Delta_{\tau(w)}(K)$ has the same Betti numbers as K .
- $\Delta_{\tau(w)}(K)$ is homotopy equivalent to a wedge of spheres.

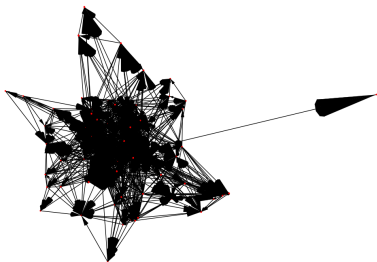
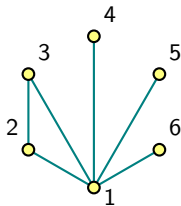


Figure: The partial shift graph starting from $\mathbb{RP}_6^2$, generated using OSCAR.

Near cones

Definition A simplicial complex K is a near cone if for every $\sigma \in K$ with $1 \notin \sigma$ we have $\sigma|_i^1 \in K$ for each $i \in \sigma$.

Lemma Björner-Kalai '88 Let K be a near cone. Then K is homotopy equivalent to a wedge of spheres and $\beta_k(K) = |\{\sigma \in K^k : \sigma \cup \{1\} \notin K\}|$.



Permutation Lemma

Lemma Björner-Kalai '88 Let $g \in \mathrm{GL}(n, \mathbb{E})$ be generic and suppose for $\tau \in \binom{[n]}{k}$ there exists $\gamma \in \mathbb{E}$ such that,

$$g_{\tau}^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma_{\sigma} g_{\sigma}^{\wedge K_k}.$$

If $\pi \in \mathrm{Sym}(n)$, then there exists $\gamma' \in \mathbb{E}$ such that

$$g_{\tau \cdot \pi}^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma'_{\sigma \cdot \pi} g_{\sigma \cdot \pi}^{\wedge K_k}$$

Shifting to near cones

Lemma DV-Joswig-Lenzen '24+ Let K be a simplicial complex on n vertices, $g \in \mathrm{GL}(n, \mathbb{E})$ a matrix with first column algebraically independent over $\mathbb{F}$ and the other entries of g . Then $\Delta_g(K)$ is a near cone.

Proof sketch

Let $\pi = (1 \ i)$ and suppose $\tau \notin K$,

$$g_{\tau}^{\wedge K_k} = \sum_{\sigma < \tau} \gamma_{\sigma} g_{\sigma}^{\wedge K_k}.$$

Then there exists $\gamma' \in \mathbb{E}$ such that,

$$g_{\tau \cdot \pi}^{\wedge K_k} = \sum_{\sigma < \tau \cdot \pi} \gamma'_{\sigma} g_{\sigma}^{\wedge K_k}$$

Rothe Matrix of the n cycle

Consider $c_n = (1\ 2 \dots n)$,

- For $1 \leq i < j < n$, $i \cdot c_n < j \cdot c_n$
- $\text{Inv } c_n = \{(i, n) : 1 \leq i < n\}$

$$\tau(c_n) = \begin{pmatrix} x_{1,n} & 1 & & \\ \vdots & & \ddots & \\ x_{n-1,n} & & & 1 \\ 1 & 0 & \dots & 0 \end{pmatrix}.$$

For $w \geq c_n$

- $\text{Inv } c_n \subset \text{Inv } w$
- $\tau(c_n)$ and $\tau(w)$ have the same first column.
- $\Delta_{\tau(w)}(K)$ is a near cone for any K .

Preserving Betti Numbers

Lemma Bjöner-Kalai '88

$$\beta_k(K) = |K^k| - |\{\sigma \in \Delta(K^{k+1}) : 1 \in \sigma\}| - |\{\sigma \in \Delta(K^k) : 1 \in \sigma\}| .$$

Proof sketch of shifting by $w \geq c_n$ preserves Betti numbers

It is enough to show,

$$|\{\sigma \in \Delta_{\tau(w_0)}(K^k) : 1 \in \sigma\}| = |\{\sigma \in \Delta_{\tau(w)}(K^k) : 1 \in \sigma\}|.$$

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$$\tau(w_0)^{\wedge S}, (\tau(w)\tau(w^{-1}w_0)^w)^{\wedge S}$$

have the same column matroid.

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Hence $(w_0w^{-1})^{\wedge k}$ fixes the set $J = \{\sigma \in \binom{[n]}{k} \mid 1 \in \sigma\}$.

Open Questions

- For what other $\mathcal{F}$ is $\Gamma_{\mathcal{F}}$ still acyclic?
- When can we determine apriori that $\Delta_{\tau(w_0)}(K) = \Delta_{\tau(w)}(K)$?
- Can we find bounded sequences $(w_i)_{i=1}^m$ such that $\Delta_{w_m}(\dots \Delta_{w_1}(K)) = \Delta(K)$

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Thank You!